Artificial Intelligence

BS (CS) Spring 2026

Lab 02 Tasks



Learning Objectives:

1. Lambda Functions and Regular Expressions
2. Introduction to Numpy
3. Introduction to Pandas
4. Matplotlib

Lab Tasks

Submission Instructions

1. Create a notebook file for all questions with name *ROLLNO_SEC_LAB01* **e.g. i23XXXX_A_LAB02**
2. Compress your **.ipynb** file along with output **screenshots** directly into a **.zip file**.
3. Now you have to submit this zipped file on Google Classroom under your section heading.
4. If you don't follow the above-mentioned submission instruction, you will be marked **zero.**

Plagiarism in the Lab Task will result in **zero** marks in the whole category.

Q1. Given the dataset students_score_data.csv you have to perform:

- **Load the Data:**
  - Create a Pandas DataFrame with the above data.
  - Display the first few rows.
- **Fill the missing values:**
  - Fill the missing values with the average value of each column
  - Remove the row where all three subjects' marks are missing
  - If the score in any subject is less than zero replace it with zero.
- **Calculate the Average Score:**
  - Create a new column Average_Score that contains the average score of Math, Science, and English for each student.
- **Determine Pass/Fail Status:**
  - Add another column Pass_Status. A student is considered "Pass" if their Average_Score is 75 or above, otherwise "Fail".
  - Encode the Pass_Status column, 0 for Pass and 1 for Fail.
  - Count the total number of students pass and calculate the percentage of pass students.
- **Determine top 3 students:**

    - Determine the top 3 based on total marks
- **Display the Updated Data:**
    - Show the DataFrame with the new columns.

Q2. Plot a scatter plot comparing Math vs. Science scores,

- where:
    - Excellent: English >= 90
    - Good: 75 <= English < 90
    - Needs Improvement: English < 75
- The size of the points is proportional to the English scores.
- Color the points based on their performance category ("Excellent," "Good," or "Needs Improvement").

Q3. Perform the following operations:

- **Create the Array:**
    - Generate a 6x6 matrix of random integers between 0 and 100.
- **Row/Column Calculations:**
    - Compute the mean, median, and standard deviation for each student (row).
    - Compute the mean, max, and min for each subject (column).
- **Identify Outliers:**
    - Replace outlier scores (those beyond 2 standard deviations from the mean of each subject) with -1.
- **Masked Array:**
    - Select students who scored above 90 in at least one subject and calculate their average score.
- **Top n Students:**
    - Implement a function to return the indices of the top n students based on their total score across all subjects.

Q4. Write a program that:

- Uses regex to check password strength
- A strong password must have following properties:
    - Be at least 8 characters
    - Contain at least one uppercase letter, and at least one lowercase letter, special character, and at least one digit.

- Print whether each password is **Strong** or **Weak**.